0.1 Inner Comparison

The cross-method and matched-fit results establish external effectiveness and isolate the contribution of teacher-conditioned deflection. We next hold the backbone, rendered inputs, pseudo labels, and evaluation protocol fixed, and test three narrower claims: whether the native numeric channel remains limiting under stronger prompting controls; whether the gain concentrates in teacher–human disagreement regimes; and whether the learned context-attention pattern is functionally responsible for the score. Exact numerical values are retained in the appendix tables.

0.1.1 Native Scoring Channel

All prompting controls use the same frozen backbone, multimodal input, and evaluation rubric. They differ only in how the native language channel is elicited. Figure 1 separates the three quantities needed to diagnose a channel bottleneck: human alignment, score resolution, and sensitivity to prompt variation.

Figure 1(a) shows that analyze-then-score is the strongest native-channel baseline at 57.6% pairwise accuracy, whereas the plain frozen readout reaches 66.8% and nuGuidance reaches 70.3%. The 12.7-point gap from the strongest prompting control is therefore not explained by a weak direct prompt. Figures 1(b–c) show the same separation in the output interface itself: nuGuidance exposes 341 effective score levels, 14.4× the 23.7 levels of the best prompting control, while reducing prompt sensitivity from 0.62 to 0.09, an 85.5% reduction. Better prompting improves formatting, but it does not recover the accuracy, resolution, or stability of a learned readout.

0.1.2 Semantic Disagreement Regimes

Overall MAE can conceal whether an evaluator succeeds on the cases that motivate semantic scoring. We therefore partition WOD-E2E using the fixed Scheme-B teacher score, public rule outputs, and existing human ratings, with no model-dependent manual case selection.

Figure 2(a–b) distinguishes semantic correction from ordinary teacher fitting. On the agreement subset, nuGuidance remains close to the teacher (1.61 versus 1.58 MAE), so its gain is not obtained by perturbing cases on which the teacher is already aligned. The advantage instead appears where context is necessary: rule over-penalty MAE falls from 3.42 for the teacher to 2.42, and rule-pass blind-spot MAE falls from 3.95 to 2.79. The plain readout remains at 2.88 and 3.41, respectively, showing that pseudo-score distillation alone does not reproduce the hard-case gain. Figure 2(c) gives the direct teacher-inheritance test: nuGuidance reaches 60.7% TranscendRate, 8.3 percentage points above the plain readout and 4.6 points above LoRA. This supports, but does not assume, weak-to-strong behaviour on the teacher's identifiable failure regime.

0.1.3 Causal Readout Interventions

We intervene on the exact context-attention branch used by the filtering criterion. One trained nuGuidance checkpoint and scalar head are held fixed, and no intervention is followed by retraining. Figure 3 reports changes relative to the unchanged readout, which is the relevant paired causal comparison.

The intervention ordering is consistent across the external metric and the internal statistic. Uniform attention, which removes all selectivity, causes the largest overall degradation. Among targeted removals, suppressing the highest-attended context mass costs 8.5 accuracy points, whereas removing equal attention mass at random costs only 1.4 points. The resulting 7.1-point gap is the decisive mechanism test: the score depends specifically on the evidence ranked highly by the learned readout, not merely on retaining any equal amount of context. The corresponding deflection drops are 6.1 and 0.8, a 5.3-point gap in the same direction. Because the checkpoint and scalar head never change, the matched degradation of external performance and ρ_{TCD} supports a functional role for the learned context-attention pattern rather than a post-hoc attention visualization.

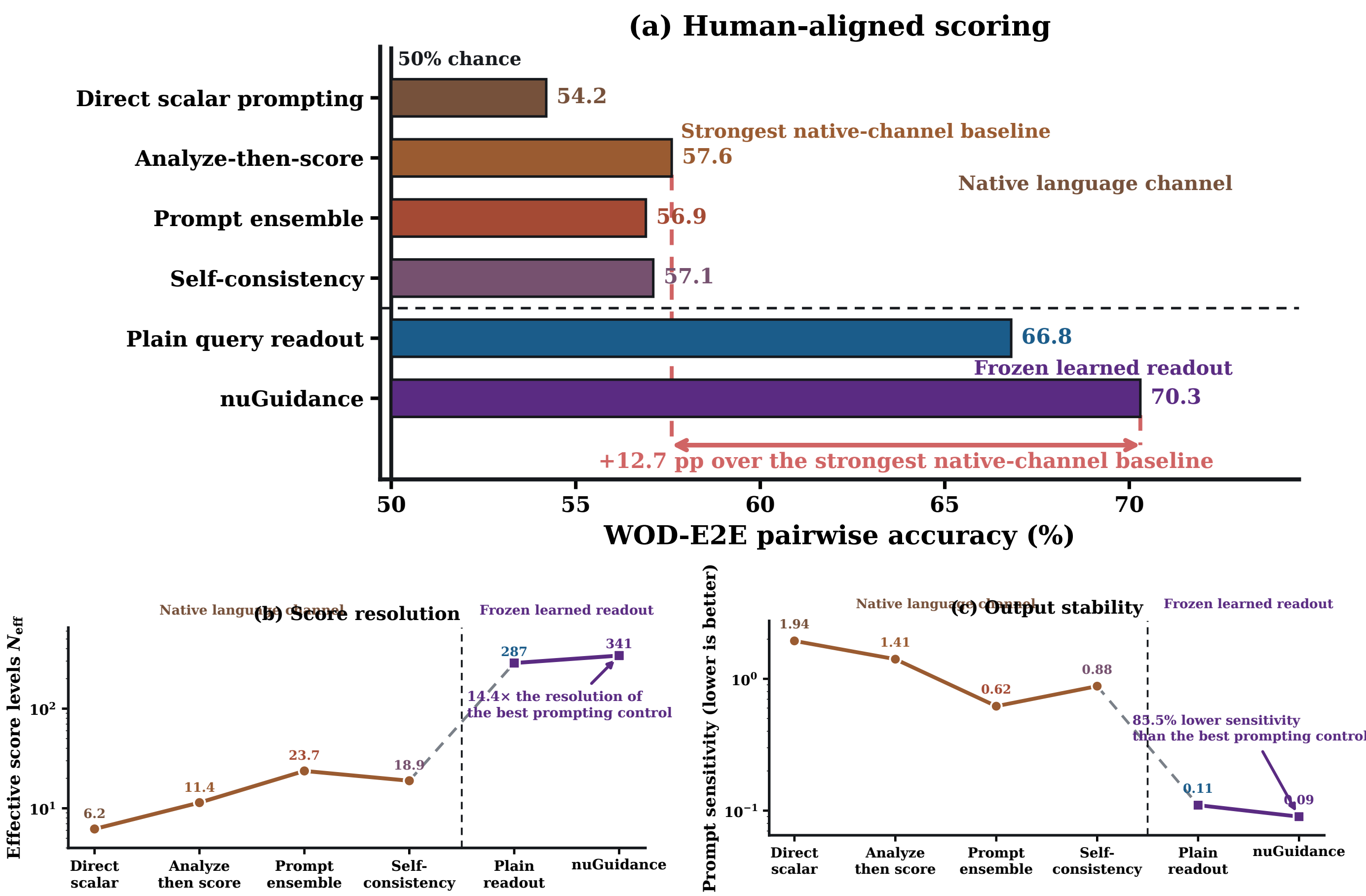

Figure 1: **Native textual scoring versus frozen learned readout.** (a) WOD-E2E pairwise accuracy, shown relative to the 50% chance reference. The horizontal annotation reports the gain over the strongest native-channel baseline. (b) Effective score resolution N_{eff} on a logarithmic scale. (c) Prompt sensitivity on a logarithmic scale, where lower is better. Stronger prompting, ensembling, and repeated sampling remain below learned readouts in all three views.

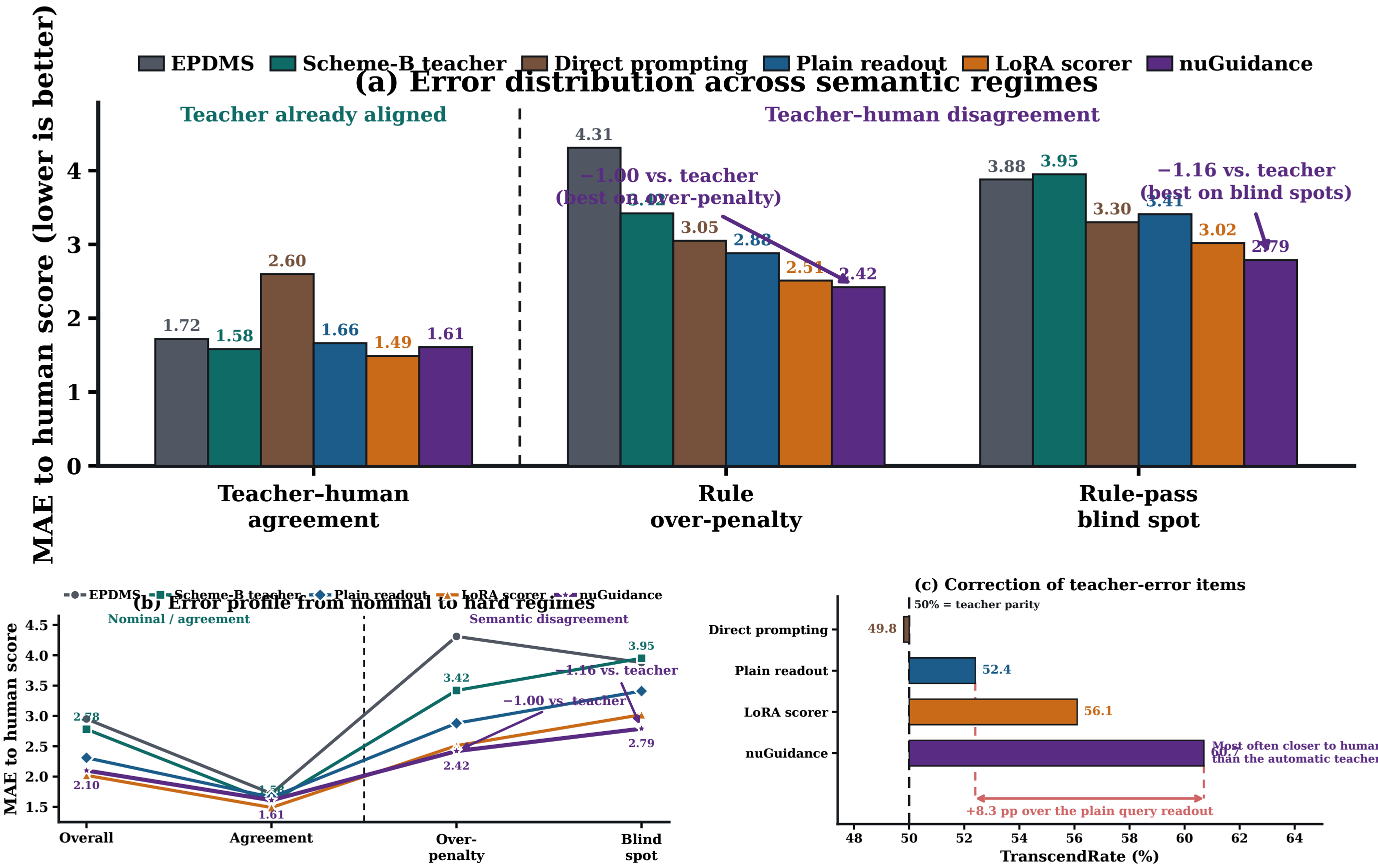

Figure 2: **Performance under semantic disagreement.** (a) Fixed-scale MAE to human ratings on teacher–human agreement, rule over-penalty, and rule-pass blind-spot subsets. (b) The same methods traced from overall error to increasingly difficult semantic regimes. (c) TranscendRate shown relative to the 50% teacher-parity reference; the horizontal annotation reports the gain over the plain query readout.

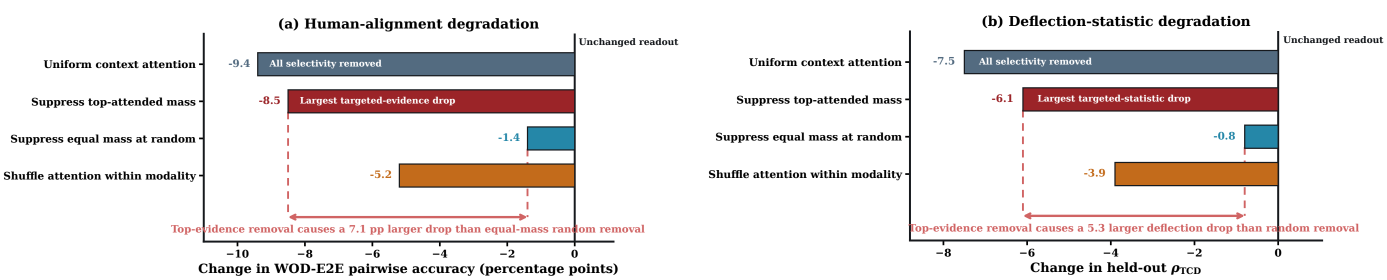

Figure 3: **Causal interventions on the learned attention readout.** (a) Change in WOD-E2E pairwise accuracy relative to the unchanged checkpoint. (b) Change in held-out ρ_{TCD}. The dashed comparison annotations expose the decisive top-attended versus equal-mass-random contrast rather than leaving the reader to subtract bar endpoints.